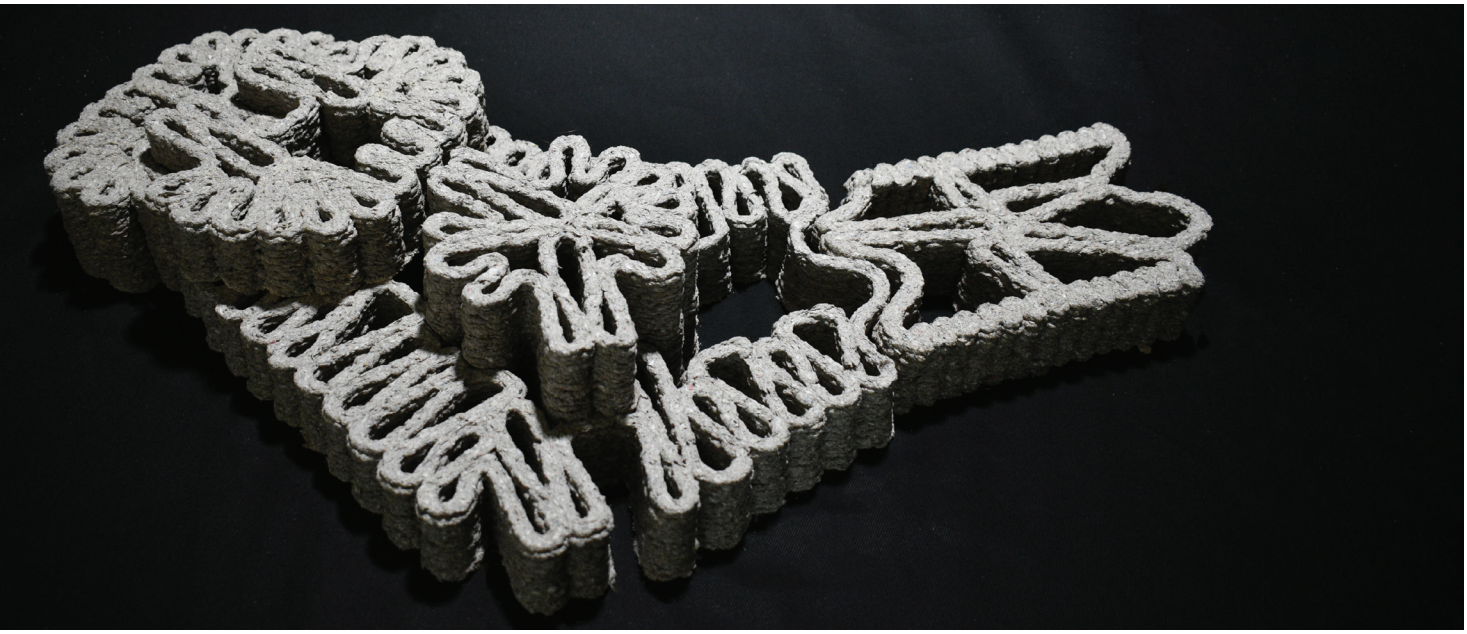
# A Material Monitoring Framework

## Tracking the curing of 3d printed cellulose-based biopolymers

Gabriella Rossi
Ruxandra Chiujdea
Claudia Colmo
Chada ElAlami
Paul Nicholas
Martin Tamke
Mette Ramsgaard Thomsen

CITA/Royal Danish Academy



1 Behavior control of cellulose-based biopolymer 3d printed components offers potential for architectural applications

## ABSTRACT

Through 3d printing, cellulose-based biopolymers undergo a two-staged hybrid fabrication process, where initial rapid forming is followed by a slower secondary stage of curing. During this curing large quantities of water are evaporated from the material which results in anisotropic deformations. In order to harness the potential of 3d printing biopolymers for architectural applications, it is necessary to understand this extended timeline of material activity and its implications on critical architectural factors related to overall element shrinkage, positional change of joints, and overall assembly tolerance. This paper presents a flexible multi-modal sensing framework for the understanding of complex material behavior of 3d printed cellulose biopolymers during their transient curing process.

We report on the building of a Sensor Rig that, using a mix of image-based, marker-based, and pin-based protocols for data collection, interfaces multiple aspects of the curing of our cellulose-slurry print experiments. Our method uses timestamps as a common parameter to interface various modes of curing monitoring through multi-dimensional time slices. In this way, we are able to uncover underlying correlations and affects between the different phenomena occuring during curing. We report on the developed data pipelines enabling the Monitoring Framework and its associated software and hardware implementation. Through graphical Exploratory Data Analysis (EDA) of three print experiments, we demonstrate that geometry is the main driver for behavior control. This finding is key to future architectural-scale explorations.

## INTRODUCTION

Industrialization has focused architectural production on materials, extracted from the geosphere, that present standardized behaviors and hold poor end-of-life possibilities. Defining ways of working with biospheric materials that are inherently regenerative and biodegradable are important steps for future sustainable building practice. Cellulose, our material base, and one of many abundant, inexpensive and renewable bio-based materials (Pattinson and Hart 2017) can be mixed with different binders, fiber reinforcement, and fillers into 3d printable slurries. However, 3d printed biopolymers undergo radical changes in their material performance and geometry as they transition from the wet to dried state. This paper reports on the Predicting Response research project investigating how the inherent heterogeneity, temporal plasticity, and anisotropy of biopolymers can be characterized in order to build advanced computational models that predict their behavior in turn enabling their usage as materials for the built environment. This paper reports on the development of the methodological steppingstones to enable such modelling practices: the making of a Monitoring Framework and Sensing Rig to characterize material behavior during fabrication.

### Data-driven Understanding of Biomaterials

Technological advancements increasingly allow architects to interface fabrication and design to the material properties of their artefacts (Ramsgaard Thomsen 2012). By combining this material design approach to new methods of data collection, we can develop new forms of analysis and modelling (Del Signore et al. 2021). Sensor-based data-driven principles in computational design research have focused on three information-rich domains where data gathering is tractable. The first is in urban and city studies where big data is gathered using city-scale sensor networks or live satellite images (Kang et al. 2020; Najari et al. 2020). The second is cyber-physical robotic fabrication, where sensors are used to measure real-time parameters punctually or via the cloud and adapt the robotic motion on the fly (Brugnaro et al. 2016; Vasey et al. 2018, Rossi and Nicholas 2018; Nicholas et al. 2020). The third is the case of monitoring material behavior, where time-based thermal sensor measurements of both samples and their environment as well as image-based captures to monitor visual properties are registered (Faircloth et al. 2018a).

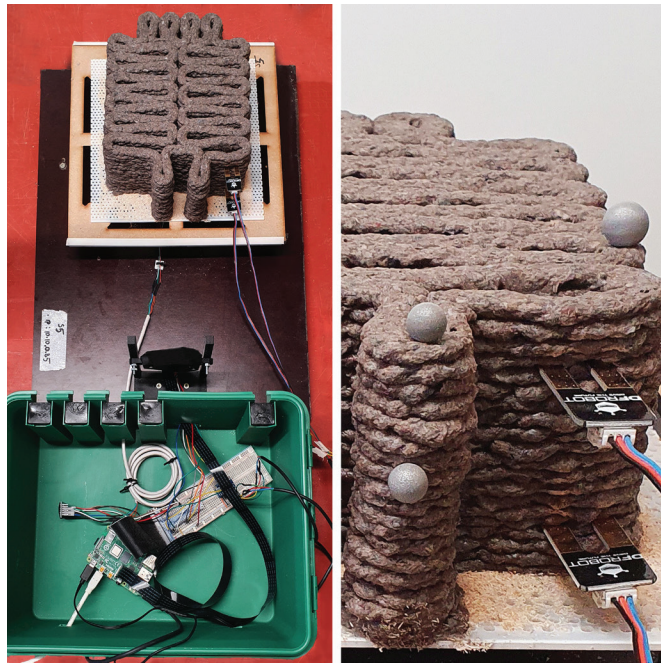
Our research sits within a growing interest in additive manufacturing of biopolymer composites. While casting might provide a high-volume production method of modular elements, 3d printing allows for a larger design space not only through geometrical freedom which has performance and formal drivers (Goidea et al. 2020), but also allow for material recipe grading (Mogas-Soldevila et al. 2015), and on the fly online adaptation. However, this potential does not come without complex manufacturing considerations: 3d printing of cellulose-based slurries is a two-staged hybrid fabrication system presenting an initial rapid forming, followed by a slower curing and hardening phase. The curing phase is characterized by the evaporation of the slurry's water content, allowing the slurry to acquire strength properties but resulting in large-scale shrinkage and warpage of the printed geometry. While geometric considerations and manipulations of curvature and edge condition matching (Goidea et al. 2020) or corrective models that overcompensate the printing toolpath to obtain as close dried target geometry as possible can be used to obtain consistent printed results (Dritsas et al. 2018; Sanandiya et al. 2018), these applications limit their investigation of the element to a binary initial "wet" and final "dry" state. We go beyond the state of the art by studying the transient curing process at different scales and across different phenomena using multidimensional feature vectors across time. By using a multi-modal sensing regime on this extended timeline of material activity, our aim is to better characterize the complexity and interactions between the curing environment and the resulting transformation thereby enabling a stronger control of the behavior of slurry biomaterials (Ramsgaard Thomsen et al. 2021), a more precise component assembly and paving the way for predictive behaviour models using Machine Learning algorithms.

In this paper we present a Material Monitoring Framework (Figure 2) for tracking the curing of 3d printed cellulose biopolymer. We first contextualize our sensing approach with respect to industry state-of-the-art curing monitoring techniques. We describe the development of our framework using IOT: a Sensor Rig which combines multiple sensors, markers and image devices, as well as data management protocols and cloud-based data storage. Our method utilizes the timestamp to bring together various modes of curing monitoring and uncover underlying correlations and affects between different phenomena, thanks to the multiple data point structure of time slices. Using Exploratory Data Analysis (EDA), we report on three printing experiments and showcase how within a controlled curing environment, geometry is the main driver for the complex behavior of cellulose-slurry curing. This finding, central to architectural and design consideration, will be our main guide for the next steps of our research.

## CONTEMPORARY SENSING PRACTICE

Architectural Sensor-Material practice is deeply embedded

in contemporary construction contexts. Powered by the rise of IOT, sensors are used for punctual quality control in pre-fabrication (Stavropoulos et al. 2013) where they are specifically designed for low-variety large-volume sensing. Another usage of sensors is lifetime tracking and safety monitoring of completed structures (Zhang et al. 2021) or of construction processes such as concrete casting. Commercially available systems, such as Sensohive's concrete monitoring [1] or the Built2spec [2] sensors can measure temperature, humidity, and $CO^2$ concentration. To give insights about the curing state. These products are limited in terms of high costs, and data analysis and synthesis customization. They are often sacrificial to the structure they monitor and cannot be reused. A low-cost cloud-based plug-and-play sensor network for architectural application has been developed by KieranTimberlake named "Pointelist" [3]. It is employed to sense the building envelope giving a real time feedback loop of the indoor level of comfort. Composed of a Wi-Fi integrated node and a Raspberry Pi microcomputer, the system uploads data from different humidity and temperature sensors, and thermal imaging to a web interface where data can be analyzed. In the Building 661 case (Faircloth et al. 2018b) the data made it possible to predict the thermal behavior and the expected material properties of the non-insulated facade. This example confirms the viability of our data-driven approach for material monitoring.



2

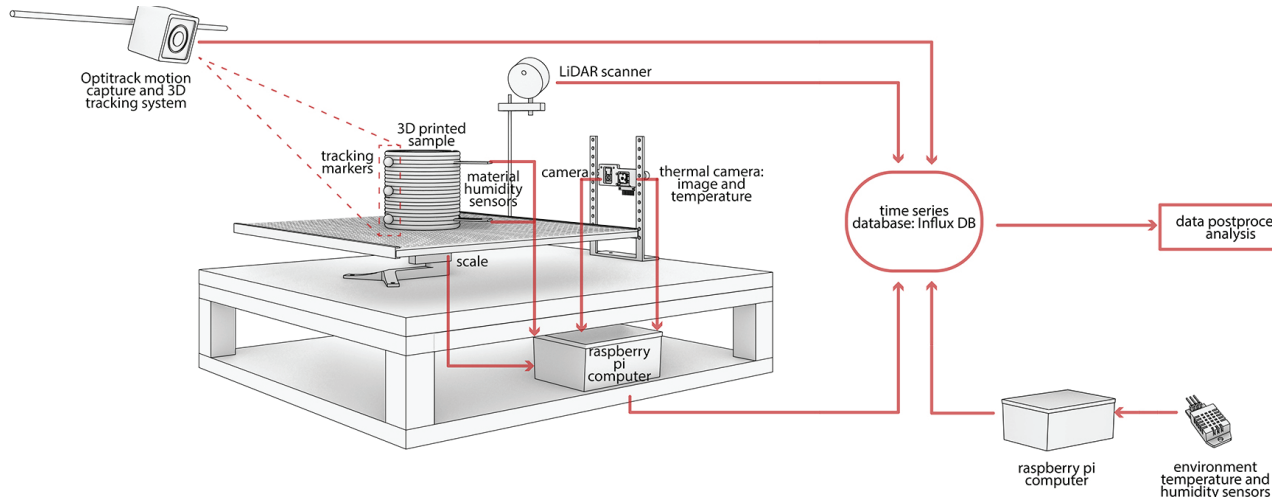
## METHOD

Studying the curing of 3d printed cellulose slurry requires monitoring of multiple phenomena across the duration of the curing process which, depending on the size of the print and the thickness of the print beads, lasts 7 to 15 days. In our research context, the Sensing Rig cannot be sacrificial but rather reusable, since we need to repeatedly monitor the curing of many material probes and analyze and compare the collected data. The Sensor Rig also needs to be replicable and able to accommodate an open-ended range of sensors. As the 3d prints are fragile when still wet, the sensors should be as non-intrusive as possible. These criteria necessitate remote and vision-based sensors. We utilize timestamp tagging as a tool to interface and compare data incoming from different sources. This creates a multiple-data-point time slice which we use to uncover data patterns and correlate phenomena driving the curing process.

### Sensing Hardware

The Monitoring Framework includes the monitoring of the following phenomena: (1) weight (sensor: 5 kg bar-type load cells HX711); (2) temperature of material surface for tracking of evaporation (sensor: measurement and false color imaging with FLIR Lepton 3.5 radiometric camera); (3) humidity in material interior (sensor: SEN00114DFROBOT resistive soil moisture sensors); (4) overall geometry (sensor: Intel RealSense LiDAR); (5) position of specific points in space (sensor: Motion capture with Optitrack PrimeX13); and (6) Texture, Color and Look (sensor: RGB imaging with PiCamera). This print local sensing is further framed by environmental sensing (sensor: relative humidity and temperature of the environment with AM2315 Sensor).

The Sensing Rig (Figure 3) sits on a board for easy repositioning. The scale is mounted using a 3d printed spacer to correct the strain. A 300 mm x 300 mm wooden frame holds the printed sample. The two camera tracking systems (piCamera and thermal camera) are fixed at 150 mm distance from the sensing plate. Material humidity sensors (SEN00114DFROBOT) are inserted into the sample to track the gradual evaporation of water. To sense changes in geometry, we have explored the possibility of integrating an Intel RealSense LiDAR scanner to record a pointcloud that can be transformed into mesh in a CAD environment; however, the usage of punctual markers to track the motion of specific points of interest proved better. We use eight Optitrack PrimeX13 cameras set up around the drying station, for this task. The cameras are calibrated using Wanding on Motive [4] to 0.1 mm tolerance. Camera feed masking ensures the stability of the reflective marker tracking by avoiding inference from reflective matter in the setup such as the humidity sensors. We use Motive's loopback streaming for marker position broadcasting. Simultaneously, we are monitoring the temperature and




3

2 Material Monitoring Framework for tracking 3d printed cellulose: closeup on our Sensing Rig; weight and moisture sensors and image and thermal cameras are connected to a RaspberryPi, while the Optitrack reflective markers are tracked by eight surrounding cameras

3 Diagram showcasing the multiple devices that constitute the Sensing Rig and the data flux between it and the cloud

humidity around the sensing station using two Raspberri Pis placed at opposite sides of the room.

### Data Pipeline

We develop custom upload and download data pipelines to fit with our custom sensing hardware setup, and to accommodate for different data types from multiple sources. We use the cloud-based service of InfluxDB [5] an open-source time-series database for storing our monitoring data. Their python-based API client [6] defines the Line protocol for data upload: a measurement is assigned to a timestamp and can be referenced using tags and fields, which allow for ease of query and analysis later. Measurements are taken every five minutes. For each experiment, an umbrella tag is created with subfields. The Optitrack motion-capture data is streamed using our custom version of the NatNet C++ client [7], and we upload *xyz* positions for every marker at 0.1 mm precision. Different measurements from the RPi sensing hardware are formatted into the Line protocol and uploaded tagged by measure, under the same experiment tag. Environment monitoring of the room temperature and humidity condition is uploaded as a separate tag. Finally, RGB color image and a false-color thermal image are time stamped and uploaded to a Google Drive storage using their Python API [8]. Once the curing is complete, the data is parsed for analysis. By using the influxDB API, we run custom queries and compile a Pandas dataframe, which allows the processing of data into meaningful information. Normalization is an important step of data processing: for instance, the timestamp is translated into curing time, the Optitrack marker coordinates are translated into distance from the initial wet point, and the weight of the sample is translated into weight loss percentage. With this method, we create a dataframe for each experiment and use graphical Exploratory Data Analysis to understand the curing process.

## CURING EXPERIMENTS SETUP

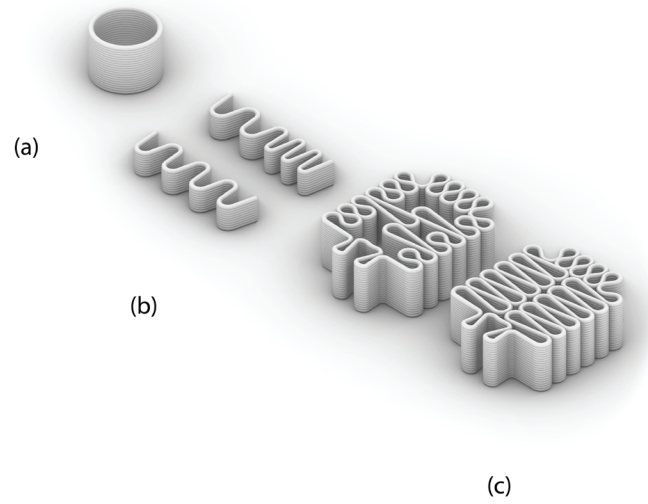
In-house variations of the cellulose slurry recipe developed at CITA (Chiujdea and Nicholas 2020) have shown the impact of varying proportions and ingredients on print behavior. The recipe used for all prints here reported optimizes nozzle flow and inter-layer adhesion, developed by our DTU partners (Rech et al. 2021a; Rech et al. 2021b). It blends cellulose floc, wood flour, glycerol, xanthan gum, calcium chloride and water (72% of the total weight). The material is packed into an acrylic tube which feeds to our custom extruder end effector. Due to the material's viscosity, a pneumatic pressure of 2 bar is sufficient to move the material to the printing head. Here a motor driven screw ensures consistent flow of material during extrusion. Using an ABB140 arm, we print at 25 mm/s speed, with a 9 mm nozzle and 4 mm layer height. The samples are printed on a perforated mesh which aids their later curing process (Figure 4).

The goal of the experiments is to isolate, for a given recipe and curing environment, the role of the print geometry in driving the curing process. We report on three prints below:

- Experiment 1 (Figure 5a) examines a cylinder as a baseline to the behavior of the material. The hypothesis



4




5

4 Print setup with ABB 140: using the perforated mesh as a print bed allows air to circulate through the sample during curing, thus avoiding molding

5 Digital printing toolpath of the three experiments; the toolpaths are spiralized to guarantee print stability

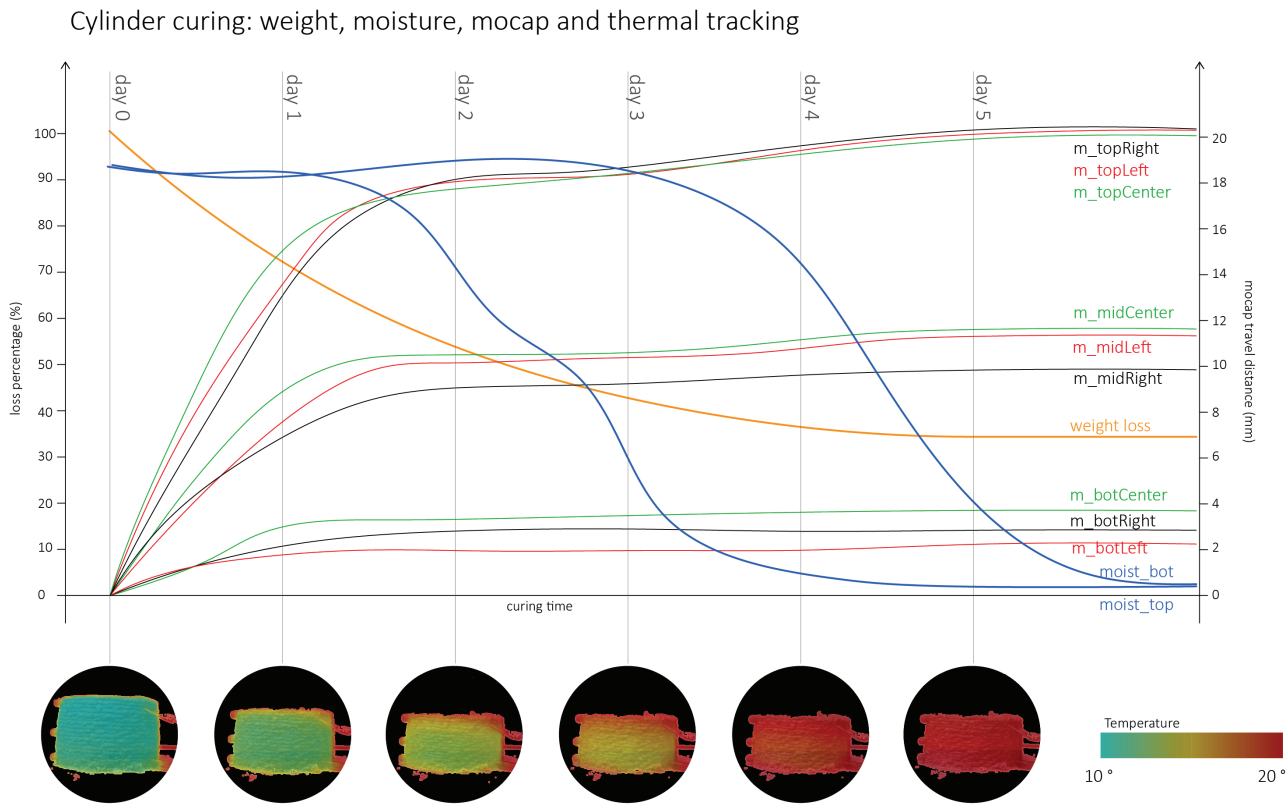
is that all material is equidistant from the center, and therefore, equally exposed to evaporation. Our sample is 100 mm in diameter and 74 mm high. It is monitored using a weight scale, two humidity sensors (one near top layer, one near bottom layer), and a 3x3 grid of Optitrack markers and thermal imaging.

- Experiment 2 (Figure 5b) aims to understand how material distribution affects evaporation, and therefore curing behavior, where larger exposure to evaporative surface would lead to faster curing. It comprises two "sine wave" samples 240 mm long: a constant 55 mm wave-length sine wave and a varying sine wave with 35 mm and 55 mm wavelengths, both 50 mm wide amplitude.
- Experiment 3 (Figure 5c) expands this understanding to volumes, with particular focus on the joints and assemblies. The hypothesis is that the density of the infill would differentiate the curing of the joint. We print two component samples of size 250 x 200 x 80mm fitted with the same male joint. Both present the same outer surface area (87,000 $mm^2$) but a varying inner evaporative surface area: a dense component with 240,000 $mm^2$ inner surface and a light component with 264,000 mm2 inner surface. The experiment questions how evaporation affects the deformation of the component and its assembly joint.

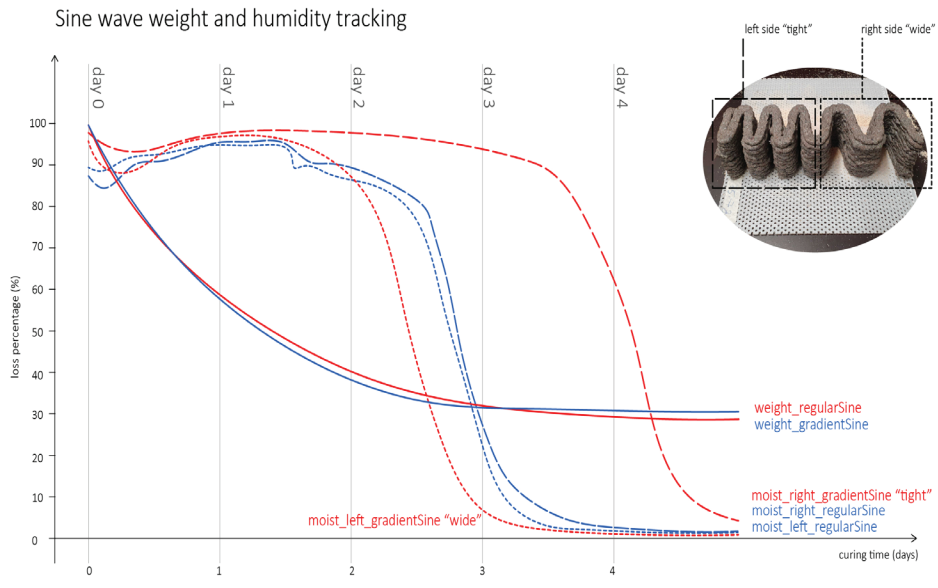
## RESULTS

### Experiment 1 "The Cylinder"

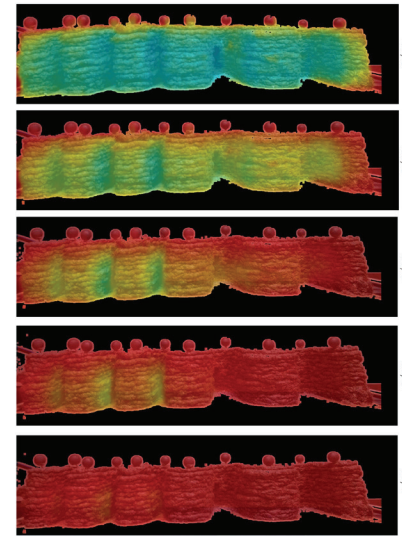
We represent the findings of our multimodal sensing through graphical Exploratory Data Analysis (EDA), see Figure 6, which shows that the cylinder print reaches a drying plateau on day 5. The sample loses approximately 60% of its weight. As water evaporates, we observe a drop in humidity registered by the moisture sensors. In the graph we can see that there is a one day delay between the sharp drop in humidity at the top sensor (at 15% weight loss) and at the bottom sensor (at 30% weight loss), indicating that the print dries from top to bottom. However, the slight increase in the bottom sensor reading indicates that water travels downwards during the print process due to gravity. This hypothesis is confirmed by the thermal imagery: the sample is initially all blue (10 degrees) and slowly transitions downwards into red as it dries and reaches room temperature (20 degrees). We also observe that 95% of the shrinkage occurs over the first 48 hours of curing. The travel distance of the Optitrack markers, as well as their travel speed is higher for the markers placed at the top of the sample (20 mm which in this case represent 30% of the total sample height), and relatively small for those placed at the bottom (approximately 2 mm). We can therefore formalize a baseline curing behavior: with all material equally exposed to an evaporative surface, a mostly vertical shrinkage occurs, with water travelling through the sample due to gravity.




6




7

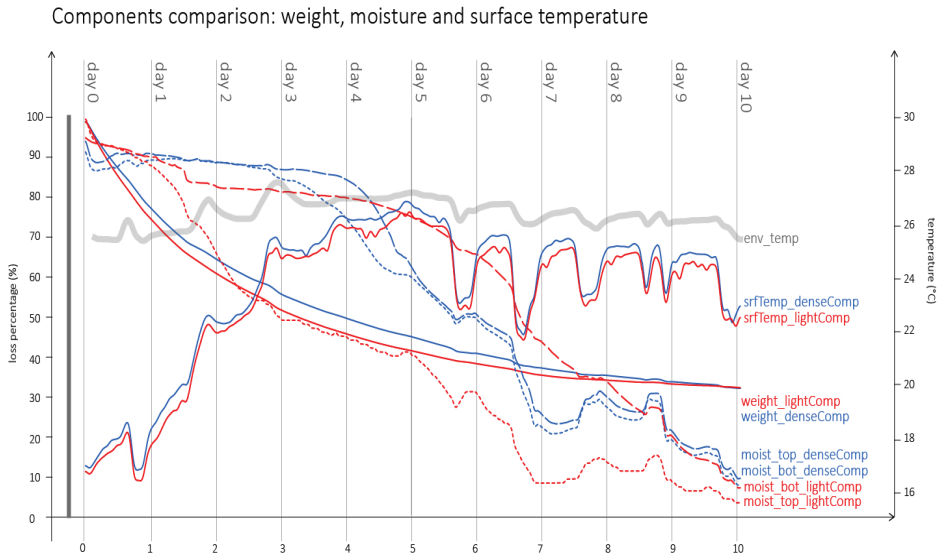



8

6 Graphical EDA of Experiment 1: the curing behaviour of a 100 mm diameter cylinder; Multi-Sensor data is overlaid in the graph, and thermo-imagery vignettes showcase the curing state every day at 13:00

7 Graphical EDA of Experiment 2: the simultaneous curing of a regular and a graded sine wave prints; the key legend shows the placement of moisture sensors of the sample

8 Monitoring of the gradient sine wave drying using thermal imaging; notice the left "tight" side of the print drying later than the right "wide" side




9

9 Graphical EDA of Experiment 3: measurements in red relate to the light component, in blue relate to the dense component, and in gray to the curing environment in which the components are left to dry

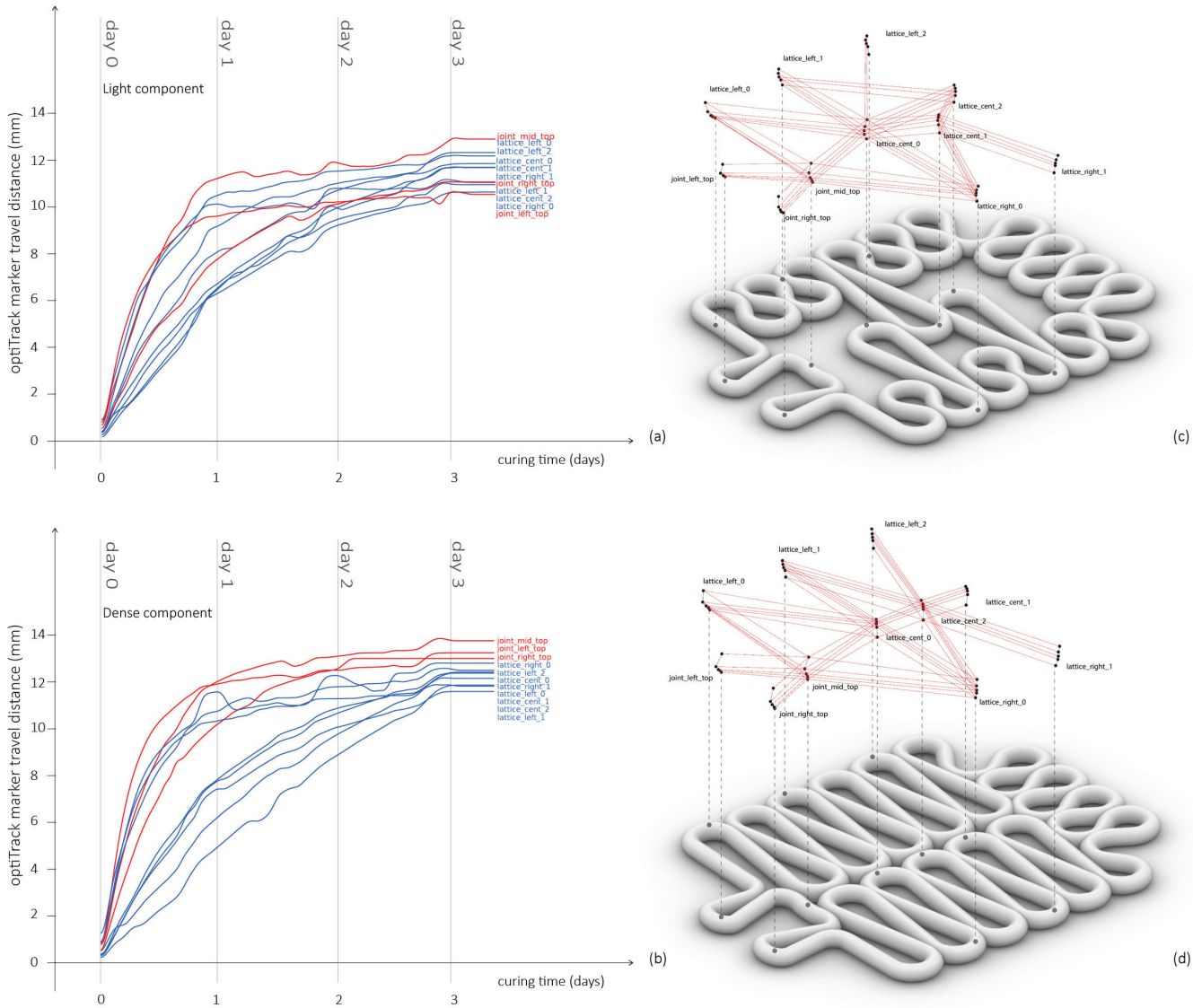
10 Plotting of the Optitrack Marker displacement across time for the light (a) and dense (b) components reveals the speed and amplitude of the shrinkage; in (c)(d) we spatially represent the geometric deformation across five curing moments; markers on the dense component (d) dry at unequal rate if compared to the ones on the light component (c)

**Experiment 2 "The Sine Wave"**

We compare the curing process of the two sine waves samples over five days (Figure 7). We observe that while both samples lose water at the same rate, the localization of this humidity loss is different. The humidity sensor readings show that in the case of the regular wave geometry, both left and right humidity sensors register a moisture loss on day 2, but in the case of the varying wave geometry, the humidity sensor on the wide wave side registers the drop early on day 2, but the one on the tight wave side only registers it on day 4. This geometry-driven differential curing behavior hypothesis is confirmed by the thermal imagery captures over the five days (Figure 8), where we see that on day 2 the wide side of the sine wave presents a surface temperature of 20 degrees—meaning it is dried—while the cavities of the tight part of the wave are still at 14 degrees—meaning they are still humid. We also observe that the vertical shrinkage is more accelerated on the wide side than on the tight side. We conclude that evaporative surface exposure—in this case the sine wave period—allows the acceleration or slow-down of the drying.

**Experiment 3 "The Modular Component"**

The two volumetric components are monitored over ten days. Through graphical EDA (Figure 9), we observe that the light component, the top moisture sensor registers a humidity drop already on day 1, while the bottom one only registers it on day 5. This is different from the dense sample, where both sensors start to register a drop between day 3 and 4. This implies that the airiness of the infill allows water to leave the material faster. Examining the minimum temperature at the joint, which has been placed facing the thermal camera, we can see that for both samples it rises steadily from 17 degrees to 24 at the same rate, while following the temperature cycles within the lab space. The central part of the joint remains colder on the dense component than the airy one. We also observe that the dense sample presents less fluctuations. We assume that this is because it has 30% more material mass and can therefore better retain the heat. Analyzing the Optitrack markers via graphical EDA (Figure 10 a/b) shows that drying does not occur equally across the samples. While Experiments 1 and 2 dried explicitly from top to bottom, in Experiment 3 we observe that the denser the infill, the more the drying will also be from edges to center. We can see that in the case of the light component (Figure 10a), all the marker points at the top layer start to move at the same time and at similar rates, no matter if it is the joint or the central or lateral part of the lattice. However, for the dense component (Figure 10b), it is the joint that starts shrinking first, followed by the lattice edges, and finally the lattice center starts to shrink as well. This is due to the way the infill structure forms evaporative cavities. Our flexible data pipelines allows us to visualize this data in different platforms: in addition to graphical EDA, we can also reconstruct and visualize the Optitrack travel data in Grasshopper/Rhino (Robert McNeel & Associates 2021) environment to understand the spatial implications of the curing process. In Figure 10 c/d we show the location of the Optitrack markers over 5 timestamps: 0, 500, 1000, 1500 and 4000 minutes after curing. This representation confirms that the motion of Optitrack markers, and therefore the shrinkage of the components, does not occur equi-temporally.




10

## DISCUSSION & CONCLUSION

In this paper, we present the development and testing of a Material Monitoring Workflow and Sensing Rig, which provides deeper insight into the complex curing behavior of 3d printed cellulose-based slurries. Our Sensing Rig hardware has been developed following an accessible approach that can be easily replicated using readily available components and open-source protocols. While initial setup and calibration of these components and data wrangling processes can be laborious, since sensors are prone to data outliers and spikes, overall, the data quality produced by our rig is sufficient for our research, where the change of measured data over time is more important than the absolute precision of a single data point. Our framework's decentralized architecture with local micro-computing power and reliable cloud database provides a stable base for continuous sensing of many probes, where sensing can continue, while changing numbers of rigs are added, maintained, and removed. Additionally the sensors on each rig are plug-and-play and their configuration can be tailored to specific experiments needs.

The experiments we have described, through their associated sense data, have revealed important aspects of material behavior during curing that have crucial implications on critical architectural factors driving design and fabrication, related to element shrinkage, positional changes of joints and fits which would affect a final assembly. We demonstrate that the curing process follows a specific non-linear timescale across the samples and occurs at a changing rate. Printed elements shrink in height (-25%) more than in print plane. The impact of gravity on the drying, as water moves downwards across the layers, is minor and may improve interlayer bonding. Moreover, while the drying phase extends up to ten days, the fact that nearly all water loss and shrinkage occurs within the first

three days implies that a printed object could be positioned and left to finish drying in place after this initial stage, as its geometry and density then remain relatively stable.

Our experiments have also revealed that geometry is the main driver for surface evaporation. Open geometries that expose more surface area to airflow dry more quickly and more evenly, therefore, will be less affected by warping. By intentionally grading the airiness of the sample, one can therefore dictate the drying "sequence" of the piece, and control not only the final tolerances of the piece, but also chose to dry more robust or more fragile parts before others. This can also be used strategically to control interlocking joint tolerances, which would guarantee the feasibility of an architectural assembly using these 3d printed components. Initial material characterization is currently underway, and points to the possibility of this material being used for interior partitioning, furniture scale, and thermal and sound insulation. Taken together, these insights point to ways in which a cellulose-based biopolymer could be graded, or a production environment configured, to manage, counteract, or speed the natural curing process, and reveal opportunities for steering and control that are missed by a binary 'before and after' approach. Further experiments will be made to unpack this fabrication and assembly strategy unique to two-staged hybrid fabrication processes, with focus on male/female joint warpage driven by the infil of the components, and explore design strategies that negotiate material distribution between the need for airyness versus structural stability.

Finally, from a conceptual point of view, we report that the timestamp is a successful method of combining multiple data streams from multiple sensor sources. Through minor data wrangling and normalization operations, we can use graphical EDA mapping of monitoring data to understand and theorize the behavior of these components. Formatting data as a time series also offers advantages for the next stages of this research, where we use statistical analysis tools such as Principal Component Analysis to backward engineer the design of our Sensor Rig and choice of sensed phenomena. Lastly, next stages of this research will also take advantage of the developed formatting pipeline, and build a library of curing samples over time, to interface with Machine Learning models and predict time-varying data readings on our next printed samples.

## ACKNOWLEDGEMENTS

This work is funded by Independant Research Fund Denmark (DFF) PROJECT NUMBER 9131-00034B "Predicing Response," in collaboration with Anders Egede Daugaard and Arianna Rech (Denmark Technical University) and John Harding (University of Reading).

## NOTES

All website references listed below were accessed October 2021.

1. https://maturix.com/knowledge-center/.
2. https://cordis.europa.eu/project/id/637221.
3. Kieran Timberlake, n.d. *Pointelist*. https://www.pointelist.com/.
4. Natural point. *Motive*, v2.2.0, 2019. https://optitrack.com/software/motive/.
5. InfluxData, *influxDB 2.0 Python Client,* 2019. https://github.com/influxdata/influxdb-client-python.
6. NaturalPoint, *NatNet SDK*, 2021. https://optitrack.com/software/natnet-sdk/.
7. Google, *Drive API,* V.3, 2021. https://developers.google.com/drive/api/v3/quickstart/python.

## REFERENCES

Brugnaro, G., E. Baharlou, L. Vasey, and A. Menges. 2016. "Robotic softness: An adaptive robotic fabrication process for woven structures." In *ACADIA '16: Posthuman Frontiers: Data, Designers, and Cognitive Machines; Proceedings of the 36th Annual Conference of the Association for Computer Aided Design in Architecture*. 154–163.

Chiujdea, Ruxandra, and Paul Nicholas. 2020. "Design and 3D printing methodologies for cellulose-based composite materials." In *Anthropologic: Architecture and Fabrication in the Cognitive Age; Proceedings of the 38th eCAADe Conference*, vol. 1., edited by L. Werner and D. Koering. TU Berlin, Berlin, Germany. 547–554.

Del Signore, Marcella, Nancy Diniz, and Frank Melendez. 2021. "Information Matters." In *Data, Matter, Design: Strategies in Computational Design*, edited by Frank Melendez, Nancy Diniz, and Marcella Del Signore. New York: Routledge.

Dritsas, Stylianos, Samuel E. P. Halim, Yadunund Vijay, Naresh G. Sanandiya, and Javier G. Fernandez. 2018. "Digital Fabrication with Natural Composites: Design and Development towards Sustainable Manufacturing." *Construction Robotics* 2 (1–4): 41–51. https://doi.org/10.1007/s41693-018-0011-0.

Faircloth, Billie, Ryan Welch, Yuliya Sinke, Martin Tamke, Paul Nicholas, Phil Ayres, Erica Eherenbard, and Mette Ramsgaard Thomsen. 2018. "Coupled Modeling and Monitoring of Phase Change Phenomena in Architectural Practice." In *Symposium on Simulation for Architecture & Urban Design 2018.* 81–88.

Faircloth, Billie, Ryan Welch, Martin Tamke, Paul Nicholas, Phil Ayres, Yulia Sinke, Brandon Cuffy, and Mette Ramsgaard Thomsen. 2018. "Multiscale Modeling Frameworks for Architecture: Designing the Unseen and Invisible with Phase Change Materials." *International Journal of Architectural Computing* 16 (2): 104–22.

Goidea, Ana, Dimitrios Floudas, and David Andréen. 2020. "Pulp Faction: 3d Printed Material Assemblies through Microbial Biotransformation." In *Fabricate 2020.* London: UCL Press. 42–49.

Kang, Yuhao, Fan Zhang, Wenzhe Peng, Song Gao, Jinmeng Rao, Fabio Duarte, and Carlo Ratti. 2020. "Understanding House Price Appreciation Using Multi-Source Big Geo-Data and Machine Learning." *Land Use Policy* 111: 104919. https://doi.org/10.1016/j.landusepol.2020.104919.

Mogas-Soldevila, L., J. Duro-Royo, D. Lizardo, M. Kayser, S. Sharma, S. Keating, J. Klein, C. Inamura, and N. Oxman. 2015. "Designing the Ocean Pavilion: Biomaterial Templating of Structural, Manufacturing, and Environmental Performance," In *Proceedings of IASS 2015 Amsterdam Symposium: Future Visions – Digital Architecture and Design*. 1–13.

Najari, Arman, Diego Pajarito, and Areti Markopoulou. 2020. "Data Modeling of Cities, a Machine Learning Application." In *SimAUD '20: Proceedings of the 11th Annual Symposium on Simulation for Architecture and Urban Design*. 1–8.

Nicholas, Paul, Gabriella Rossi, Ella Williams, Michael Bennett, and Tim Schork. 2020. "Integrating Real-Time Multi-Resolution Scanning and Machine Learning for Conformal Robotic 3D Printing in Architecture." *International Journal of Architectural Computing* 18 (4): 371–84. https://doi.org/10.1177/1478077120948203.

Pattinson, Sebastian W., and A. John Hart. 2017. "Additive Manufacturing of Cellulosic Materials with Robust Mechanics and Antimicrobial Functionality." *Advanced Materials Technologies* 2 (4): 1600084. https://doi.org/10.1002/admt.201600084.

Ramsgaard Thomsen, Mette ,and Ayelet Karmon. 2012. "Informing Material Specification." In *Digital Aptitudes: + Other Openings*. Boston: ACSA. 677–83.

Ramsgaard Thomsen, Mette, Paul Nicholas, Martin Tamke, and Tom Svilans. 2021. "A New Material Vision." In *Data, Matter, Design: Strategies in Computational Design*, edited by Frank Melendez, Nancy Diniz, and Marcella Del Signore. New York: Routledge.

Rech, Arianna, Paul Nicholas, Mette Ramsgaard Thomsen, and Anders E. Dougaard. 2021a. "Waste-based biopolymer slurry for 3d printing with recycled materials." In *Nordic Polymer Days 2021*. https://nrs.blob.core.windows.net/pdfs/nrc-2-41381210-885e-4e8b-9ea3-054d3ec822b4.pdf.

Rech, Arianna, Ruxandra Chiujdea, Claudia Colmo, Gabriella Rossi, Paul Nicholas, Martin Tamke, Mette Ramsgaard Thomsen, and Anders Egede Daugaard. 2021b. "Predicting Response: Waste-based biopolymer slurry recipe for 3d-printing (CelluloseFloc_v01)." Data set. Zenodo. https://doi.org/10.5281/zenodo.5557218.

Robert McNeel & Associates, *Rhinoceros 3D & Grasshopper*. V.7, SR13. Windows. 2021.

Rossi, Gabriella, and Paul Nicholas. 2018. "Modelling a Complex Fabrication System: New Design Tools for Doubly Curved Metal Surfaces Fabricated Using the English Wheel." In *Proceedings of ECAADe 2018*. 811–20.

Sanandiya, Naresh D., Yadunund Vijay, Marina Dimopoulou, Stylianos Dritsas, and Javier G. Fernandez. 2018. "Large-Scale Additive Manufacturing with Bioinspired Cellulosic Materials." *Scientific Reports* 8 (1): 8642. https://doi.org/10.1038/s41598-018-26985-2.

Stavropoulos, P., D. Chantzis, C. Doukas, A. Papacharalampopoulos, and G. Chryssolouris. 2013. "Monitoring and Control of Manufacturing Processes: A Review." *Procedia CIRP* 8: 421–25. https://doi.org/10.1016/j.procir.2013.06.127.

Vasey, Lauren, Ehsan Baharlou, Moritz Dörstelmann, Valentin Koslowski, Marshall Prado, Gundula Schieber, Achim Menges, and Jan Knippers. 2015. "Behavioral Design and Adaptive Robotic Fabrication of a Fiber Composite Compression Shell with Pneumatic Formwork." In *ACADIA '15: Computational Ecologies: Design in the Anthropocene; Proceedings of the 35th Annual Conference of the Association for Computer Aided Design in Architecture*. 297–309.

Zhang, Lixiao, Guoyang Qiu, and Zhishou Chen. 2021. "Structural Health Monitoring Methods of Cables in Cable-Stayed Bridge: A Review." *Measurement* 168 (January): 108343.

## IMAGE CREDITS

Figures 1-10: © Photographs and drawings by the authors; CITA / Royal Danish Academy

**Gabriella Rossi** is a PhD fellow at CITA.

**Ruxandra Chuijdea, Claudia Colmo, and Chada el Alami** are research assistants at CITA.

**Paul Nicholas** is Associate Professor at the Royal Danish Academy, senior researcher at CITA, and leads the international masters program Computation in Architecture.

**Martin Tamke** is Associate Professor at the Royal Danish Academy and senior researcher at CITA.

**Mette Ramsgaard Thomsen** is Professor of Digital Technology at the Royal Danish Academy and leads CITA.